

CRESTON, IA, USA

WMO: 725474

Lat: 41.020N

Lon: 94.364W

Elev: 1294

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04915

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.7	0.6	-12.6	2.9	-4.2	-8.0	3.7	1.8	31.7	23.2	28.5	24.6	8.9	300	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	91.4	76.7	90.1	75.8	87.4	74.3	79.8	88.6	78.0	85.9	76.2	84.1	11.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	148.6	85.4	75.2	138.5	83.7	73.2	129.2	81.4	44.4	88.7	42.4	85.8	40.5	84.1	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.4	24.5	21.1	-12.1	96.3	5.4	3.8	-15.9	99.0	-19.0	101.3	-22.1	103.4	-25.9	106.2		
			-12.6	81.7	5.2	2.1	-16.3	83.2	-19.4	84.4	-22.3	85.6	-26.1	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	23.8	27.0	38.8	51.0	62.1	71.2	75.3	73.0	65.4	52.6	39.5	26.9
	DBStd	20.60	12.26	12.44	11.88	9.67	8.21	6.22	5.86	5.68	8.19	9.64	10.54	11.67
	HDD50	3093	811	645	384	100	7	0	0	0	3	85	340	718
	HDD65	6201	1276	1063	814	428	153	18	4	7	95	398	765	1180
	CDD50	3337	0	2	36	130	382	635	783	712	464	166	25	2
	CDD65	970	0	0	1	8	63	202	322	255	105	14	0	0
	CDH74	8889	0	0	11	116	568	1792	3165	2182	923	129	3	0
	CDH80	3158	0	0	1	23	157	600	1275	789	289	24	0	0
Wind	WSAvg	10.3	11.1	11.1	11.9	12.8	11.3	9.6	7.9	7.3	8.6	10.2	11.0	10.6
Precipitation	PrecAvg	35.5	1.0	1.1	2.2	3.7	5.0	5.1	4.5	4.0	3.3	2.3	1.9	1.3
	PrecMax	50.7	2.0	2.6	5.6	7.4	9.7	13.1	12.3	11.1	11.7	5.8	5.1	2.8
	PrecMin	25.7	0.4	0.2	0.1	1.4	1.4	0.9	1.2	1.2	1.0	0.2	0.0	0.1
	PrecStd	7.6	0.4	0.6	1.4	1.5	2.0	3.0	2.6	2.6	2.4	1.6	1.3	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	63.7	74.9	83.6	90.1	91.2	97.1	95.3	90.5	83.6	72.6	60.6
		MCWB	45.5	52.4	59.9	64.1	72.4	75.1	76.7	76.2	73.0	66.7	58.4	52.5
	2%	DB	49.6	55.4	68.5	77.2	84.0	89.8	92.9	90.9	87.6	77.4	65.8	52.5
		MCWB	42.9	47.9	56.4	61.4	69.8	74.9	78.0	76.6	72.3	62.7	56.2	47.3
	5%	DB	44.8	49.8	63.2	72.4	80.7	86.3	90.4	88.0	82.5	72.5	60.7	47.5
		MCWB	39.6	43.3	54.1	59.3	67.1	73.4	77.0	75.2	70.2	61.4	51.5	42.5
	10%	DB	39.0	44.6	57.0	67.9	76.6	83.6	87.6	84.1	79.4	68.2	55.1	42.7
		MCWB	35.2	39.4	49.8	56.5	65.1	72.0	75.6	73.7	67.5	59.2	47.8	38.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	55.6	63.4	67.5	76.1	79.7	82.9	82.1	77.0	71.8	61.3	55.4
		MCDB	53.9	59.1	70.4	78.6	83.5	88.1	92.0	88.8	86.2	79.7	68.4	58.7
	2%	WB	43.3	48.5	59.3	64.0	71.8	77.2	80.4	79.0	74.8	67.7	57.2	48.5
		MCDB	49.5	56.6	64.8	73.9	81.1	85.3	89.9	86.9	83.5	74.1	63.4	51.1
	5%	WB	39.3	43.7	55.1	60.9	69.4	75.3	78.7	77.1	72.8	64.1	53.2	42.4
		MCDB	44.3	49.2	62.0	70.1	78.1	83.7	87.2	84.5	80.4	68.7	59.1	46.6
	10%	WB	35.8	39.1	50.1	57.8	67.0	73.7	76.9	75.1	70.4	60.3	49.0	38.4
		MCDB	39.3	43.6	57.7	65.7	74.1	81.5	84.7	82.7	76.8	67.1	55.7	42.3
Mean Daily Temperature Range	5% DB	MDBR	17.2	17.6	19.4	21.0	19.2	18.8	19.0	19.3	21.8	20.4	18.7	16.0
		MCDBR	23.4	24.6	27.6	28.0	23.3	20.8	21.4	22.8	23.9	24.7	25.6	21.6
		MCWBR	17.4	17.2	16.8	16.4	12.0	10.5	10.3	11.1	11.8	13.4	16.3	16.1
	5% WB	MCDBR	22.1	23.1	23.9	24.3	19.9	18.9	18.9	19.4	20.5	20.1	22.7	19.7
		MCWBR	17.0	17.0	15.9	15.9	11.6	11.5	11.2	11.9	11.5	14.0	16.6	15.8
Clear-Sky Solar Irradiance	taub	0.287	0.310	0.326	0.380	0.407	0.413	0.427	0.418	0.380	0.334	0.295	0.283	
	taud	2.427	2.374	2.406	2.272	2.269	2.297	2.248	2.289	2.346	2.468	2.514	2.457	
	Ebn at Noon	278	286	294	283	276	274	269	268	271	274	274	270	
	Edh at Noon	25	31	33	41	42	41	43	40	35	28	23	23	
All-Sky Solar Radiation	RadAvg	636	940	1213	1527	1806	2004	2016	1755	1477	1009	694	536	
	RadStd	57	61	109	103	146	178	150	107	103	116	57	53	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page